Chakravyuh Glucose Twin

A Type 2 Diabetes digital twin proof-of-concept

Team Chakravyuh · Digital Twin Challenge 2026

Synthetic data. Research demonstration.

A focused healthcare question

Can patient history improve short-term glucose forecasts?

Use a 3-hour observed window to forecast the next 2 hours

Support clinician review of a possible glucose excursion

The prediction contract

Eight forecasts: +15, +30, …, +120 minutes

A separate score for any reading above 180 mg/dL

An excursion may already be underway at prediction time

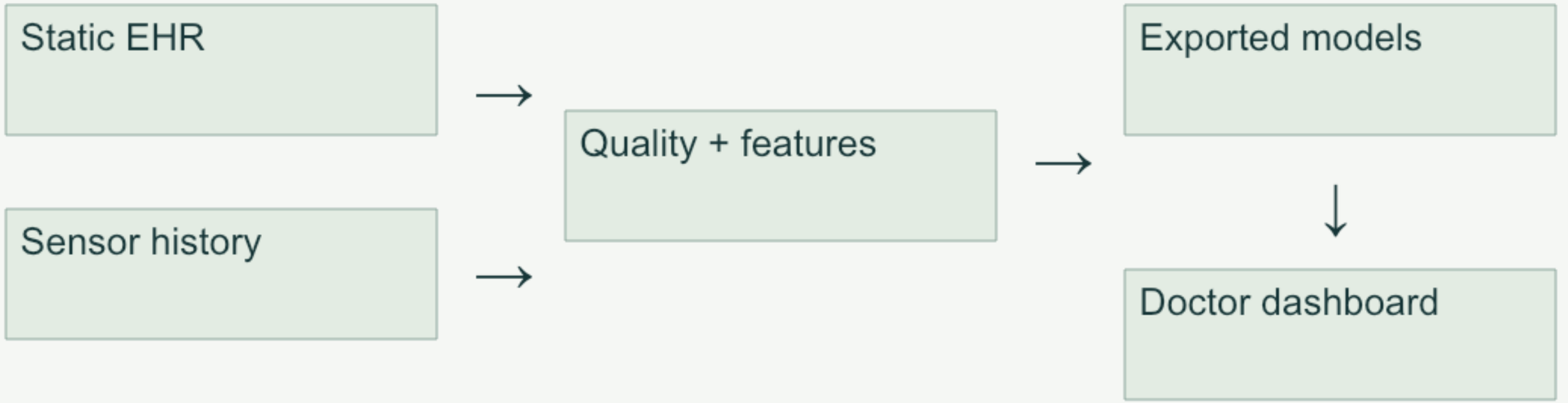
Two synthetic data streams

Static EHR: age, BMI, HbA1c, diabetes duration

Dynamic: CGM, steps, heart rate and sleep

Auxiliary meal log: carbohydrate grams already recorded

System architecture



Offline Python exports coefficients and evaluation. Runtime sees only past observations.

A patient state that updates

15 features combine EHR and recent history

A new sensor sample changes the current twin state

No future CGM values or future meals enter the features

Transparent baseline models

Multi-output ridge regression for the glucose trajectory

Regularized logistic regression for excursion risk

Training-only scaling; exported coefficients for local inference

Patient-level evaluation

54 training patients / 18 validation / 18 test

7,051 training windows / 2,311 validation / 2,303 test

Threshold and interval widths come from validation only

Synthetic regression results

Fusion 120-minute MAE: 13.205 mg/dL

Persistence baseline: 15.136 mg/dL

Patient-bootstrap 95% interval: 12.344–14.023 mg/dL

Alert limitations are substantial

Fusion detects 13 of 49 positive test windows

Recall 26.5%; F1 0.394, below sensor-only F1 0.521

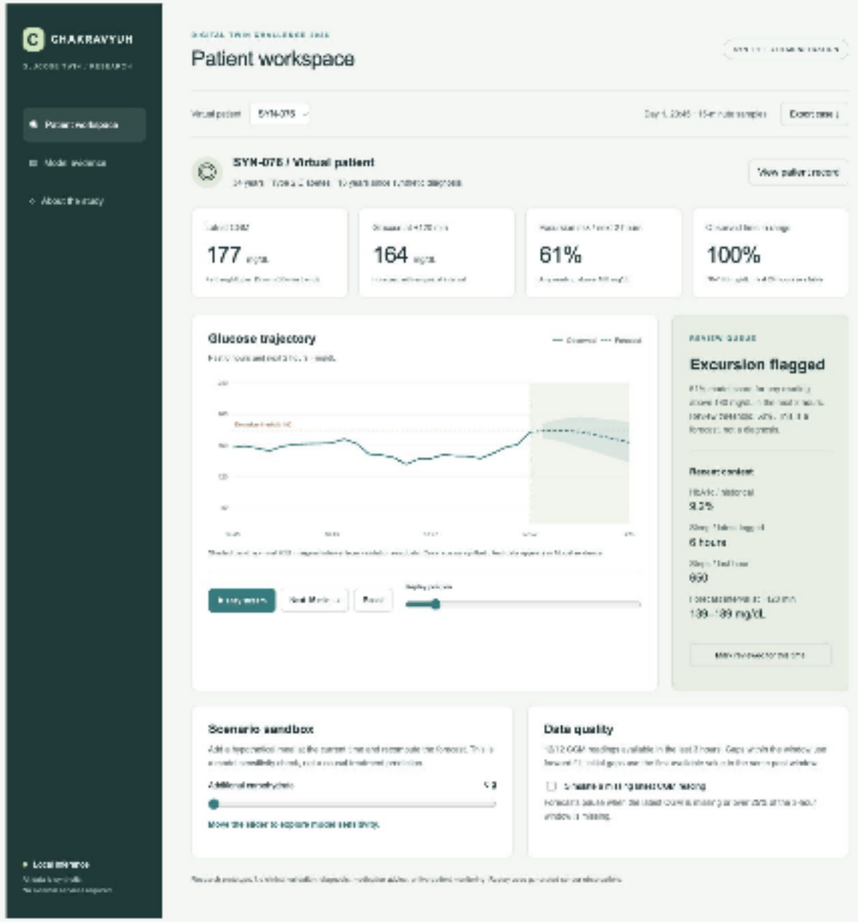
Onset-only: 2 of 32 detected; recall 6.25%

Doctor dashboard demonstration

SYN-076 at Day 1, 20:45

Current 177 mg/dL; excursion score 61%

Forecast at +120 minutes: 164 mg/dL



Patient record and provenance

Every displayed person is a generated virtual patient

Historical EHR and sensor context remain inspectable

No live hospital, device or identity integration

Replay and state updates

Play advances one simulated 15-minute sample at a time

Step and timeline controls support repeatable inspection

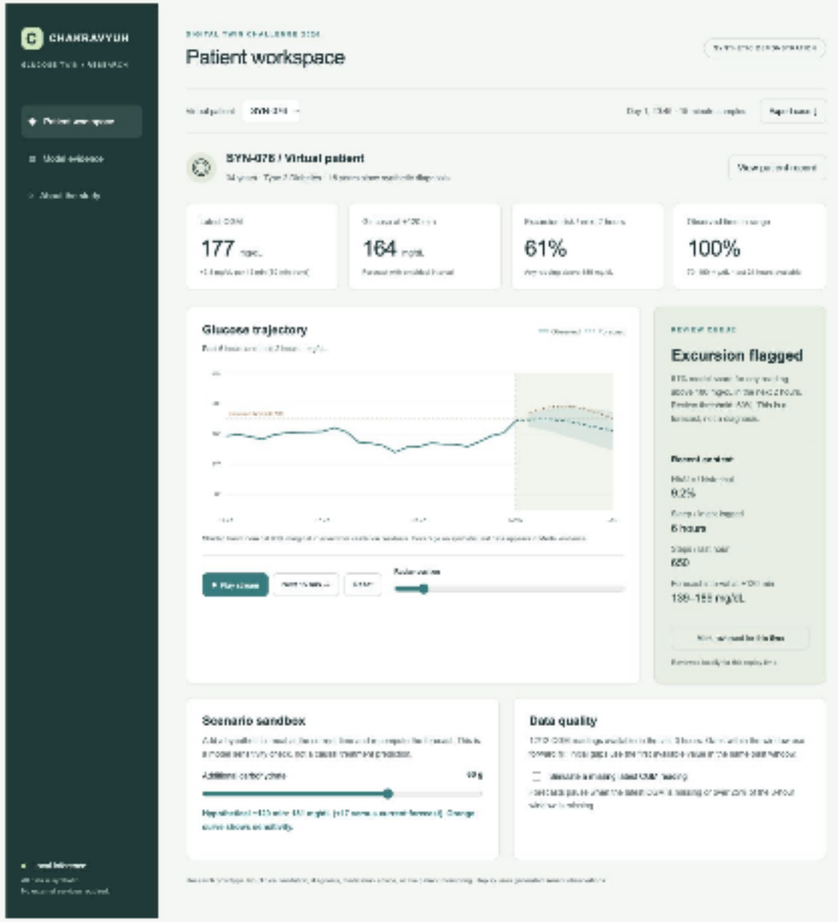
Only observations up to the cursor reach the model

Scenario sensitivity demonstration

Add 60 g of hypothetical carbohydrate at the current time

Example endpoint changes from 164 to 181 mg/dL

A feature perturbation, not a causal treatment simulation

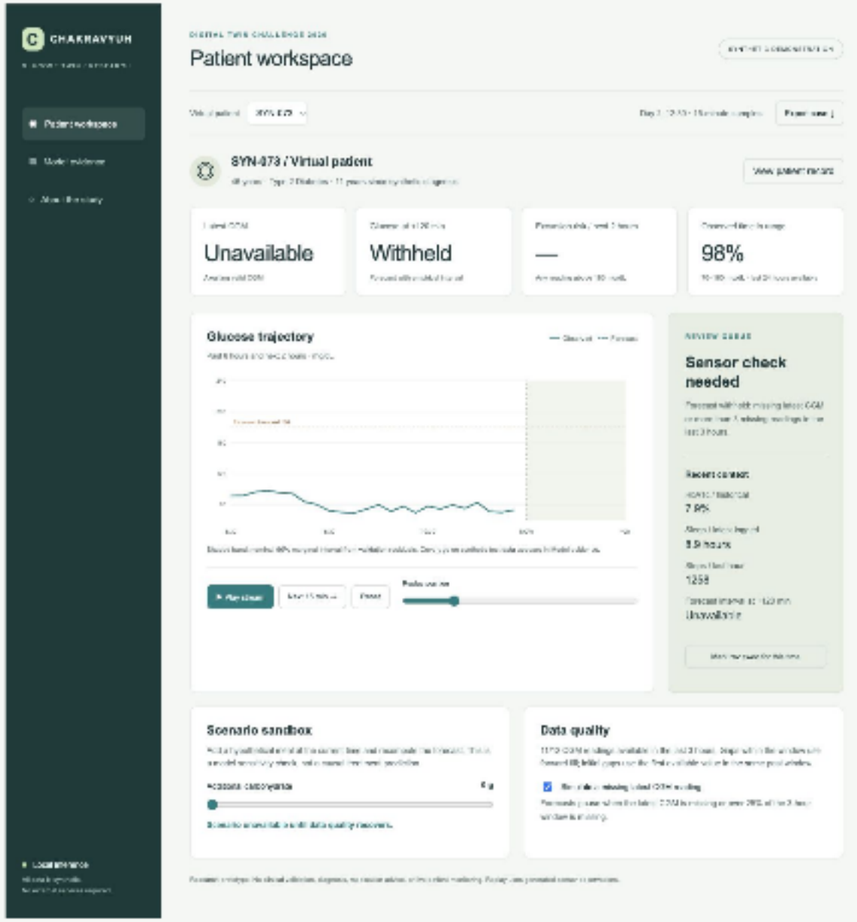


Missing data changes the response

Latest CGM missing: forecast withheld

More than 3 missing values in 12 readings: withheld

Quality status remains visible to the reviewer



Review and case export

A review marker belongs to one patient and replay time

Export includes the synthetic record and baseline inference

No data leaves the browser until the user downloads it

Reproducible code and tests

NumPy pipeline and dependency-free browser dashboard

Seven Python tests plus JavaScript feature parity

Seed, data, coefficients and metrics ship together

Uncertainty and external validity

Nominal 90% interval: 87.67% coverage at +120 min

Training and testing share one unvalidated simulator

No clinical accuracy, fairness or outcome claims

Next research milestones

Freeze this benchmark before further experiments

Stress-test onset detection and unfamiliar patient patterns

Plan licensed data access and clinician-led validation

Submission package and team handoff

Code, data, model card, architecture and presentation

Narrated demonstration exceeds the 20-minute requirement

College, member details and submission links need final verification